ECEN 248 - Lab Report

Lab #4

Simple Arithmetic Logic Unit

Section: 524

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Objectives:

The purpose of this lab is to construct and test a simple 4-bit arithmetic logic unit (ALU). This ALU will demonstrate basic arithmetic functions. This lab also introduces concepts of designing and implementing 2's complement and multiplexers. We will continue to implement DC Power Supply and bread board to test the ALU circuit. The goal of the lab is to produce an ALU circuit that can execute addition, subtraction and bitwise AND on 4-bit binary inputs. The expected results will output a 4-bit result. This lab will provide hands-on experience implementing digital circuits using a multiplexer and full adder, and further our understanding of an ALU circuit.

Design:

First, set up the power supply for the experiment. While the power supply is turned off, connect the power supply leads to the breadboard using only channel one (yellow): red cable connected to the power supply's positive terminal and the pin of this cable to the positive power rail on the breadboard, and the black cable connected to the power supply's negative terminal and the pin of this cable to the negative power rail. The power and ground was distributed across the breadboard using running wires from the top rails to the other rails located on the breadboard (this makes it easier to design the ALU circuit).

Next, set up the breadboard with a full adder, AND gate, XOR gate, and MUX, and connect the corresponding VCC and GRND lines. Then set up the dip switches, where it will be implemented with jump wires. Next, set up two 4-bit inputs for both A and B (A0, A1, A2, A3, B0, B1, B2, B3) plus two control signals C0 and C1. We placed the jumper wire on one side connected to the negative rail (for now), and the other on the circuit area of the breadboard (for each input, it has a corresponding row on the board where the other side of the jumper wire is connected).

First, we implemented the 4 bit addition/subtrction with the XOR gate and FA circuit. To do addition C1 = 0 and for subtraction C1 = 1. C0 does not affect this current circuit. The XOR gates take B inputs and C1 as inputs, when C1 = 1, it inverts be to implement 2's complement subtraction.The full adder outputs 4 sums: SUM0, SUM1, SUM2, and SUM3 and we used LED's to represent the output.
TO verify this circuit, we used simple examples like 8+5 and 8-5.

Then we connected 4 bits with an AND gate. This gate has 4 outputs: AND0 (A0B0), AND1 (A1B1), AND2 and AND3. Using the outputs from the full adder (SUM#) and AND gate, these were used as the inputs for the 2:1 MUX circuit along with C0 and GRND. If C0= 0, the AND result passes through, if C0 = 1, the ripple carry adder output passes through. The 4 bit 2:1 multiplexer will have 4 outputs: R0, R1, R2, and R3 which is visualized by LED's

Results:

Three tests were run to verify the ALU circuit. The first test was the addition of 7 + 8. C0 was set to 1 and C1 = 0 (Addition). A was set to 0111 (7), meaning that A0 = 0, A1 = 1, A2 = 1 and A3 = 1. B was set to

1000, meaning that B0 = 1, B1 = 0, B2 = 0 and B3 = 0. Inputs that are 1 are connected to the positive rail, and inputs that are 0 are connected to the negative rail. The expected result of 7 + 8 is 15, which is 1111. The LED's illuminated this expected output.

The second test was the subtraction of 7 - 8. C0 was set to 1 and C1 = 1 (Subtraction). A was set to 0111 (7) and B was set to 1000, which represents -8 in two's complement. The expected result of 7 - 8 is -1, which is a 4-bit twos complement illustrated as 1111. The output was visualized through LED's with all four of them turned on. This is the expected result/behavior.

The last test was to verify the bitwise AND operation of 7 & (8). C0 = 0. C1 does not matter in this case since C0 is not set to the math result. A was set to 0111 (7), meaning that A0 was set to the negative rail while the other bits were set to the positive rail. B was set to 1000. The expected result was 0000 and the LED's illuminated this result successfully.

Conclusion:

To conclude this lab, the 4-bit ALU was designed and tested on the breadboard successfully. The ALU performed the basic arithmetic operations: addition, subtraction, and bitwise AND, using LEDs to output the results. The ALU was constructed using a full adder, XOR gate, AND gate, and 2:1 mux by signals C0 and C1. We successfully verified the three operations: Addition (7+8 = 15), Subtraction (7 - 8 = -1) which correctly implemented the 2's complement using the XOR gates to invert the B input, and the bitwise AND (7 & 8 = 0). All the results were visualized through the LED outputs. This lab provided us with hands-on experience and furthered our understanding of ALU's, muxes, 2's complement, and 4 bit addition and subtraction.

Post-Lab Deliverables:

1. Observe and fill in Table 1. Both A and B are two's complement numbers. The result should be a 4-bit binary number. Determine whether overflow occurs or not.

The following table demonstrates the 4-bit ALU behavior across 7 test cases with different control signals C1 and C0, operations (AND, add, sub), and two's complement inputs A and B. When C0 = 0, it is always the AND operation (C1 does not matter in this case) and as we can see from the table, it produces results without overflow. An example of when overflow occurs is demonstrated in rows 3 and 5, both of these rows are performing addition (C0 = 1, C1 = 0). For row 3, overflow occurs because two positive numbers gave a negative result. For row 5, overflow occurs because two negative numbers gave a positive result. This table demonstrates that overflow happens when arithmetic operations on same-sign operands yield an opposite-sign result.

Table 1. ALU Behavior

C0	C1	Operations	A	B	Result	Overflow
0	0	AND	0100	0110	0100	No
0	1	AND	0110	1101	0100	No

1	0	Addition	0100	0110	1010	Yes
1	0	Addition	0100	1101	0001	No
1	0	Addition	1101	1001	0110	Yes
1	1	Subtraction	0100	0111	1101	No
1	1	Subtraction	0110	1001	1101	Yes

2. Determine the maximum gate delay through your final ALU circuit assuming each gate has a delay of 1 unit. Highlight the critical path on the gate-level schematic.

Maximum Gate Delay: 12 units

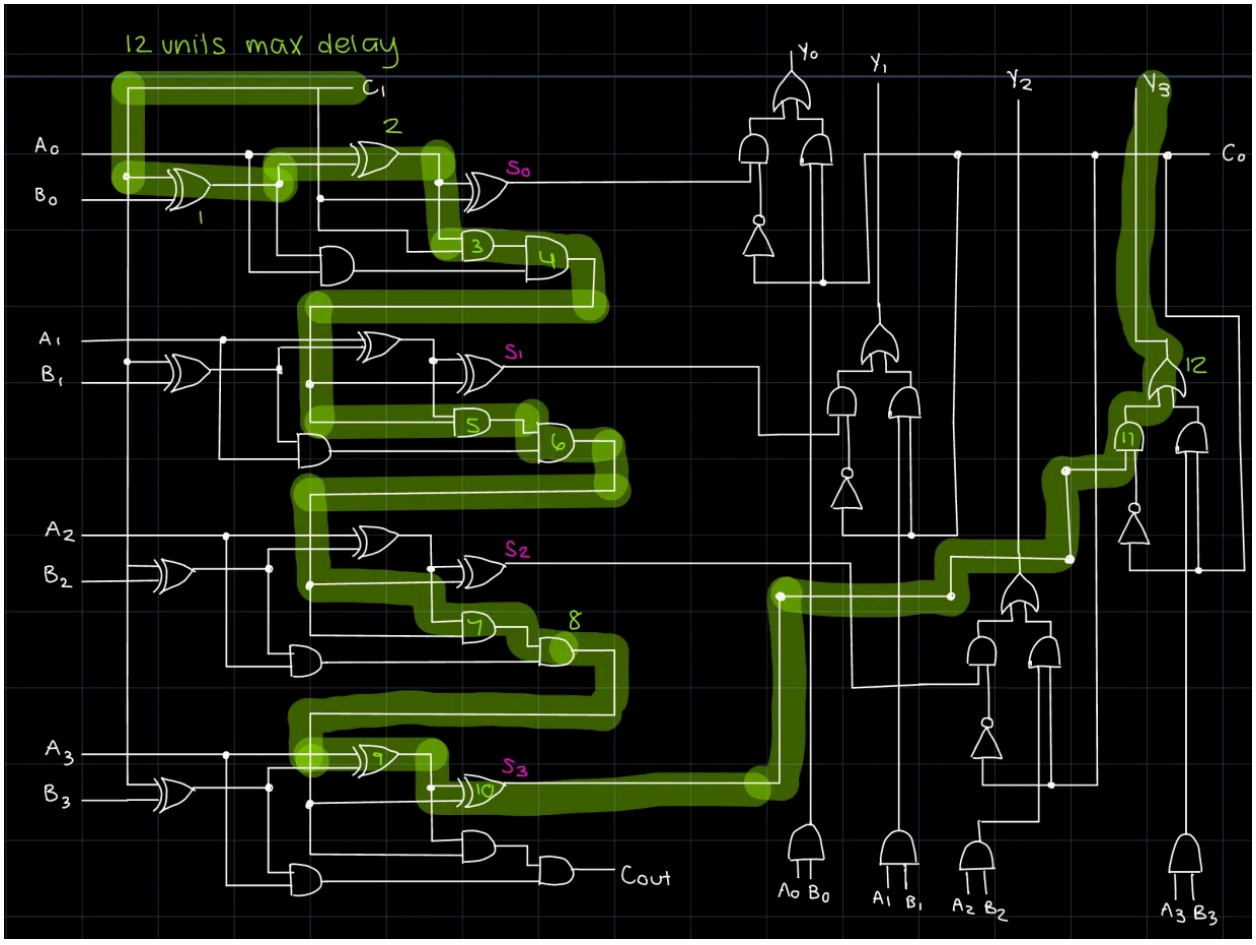


Figure 1. Critical Path and Maximum Gate Delay (12 units)

3. Please design the overflow detecting unit in Figure 6. You can use all available signals except the signals inside the chip package. Show your process and draw a gate-level

Schematic.

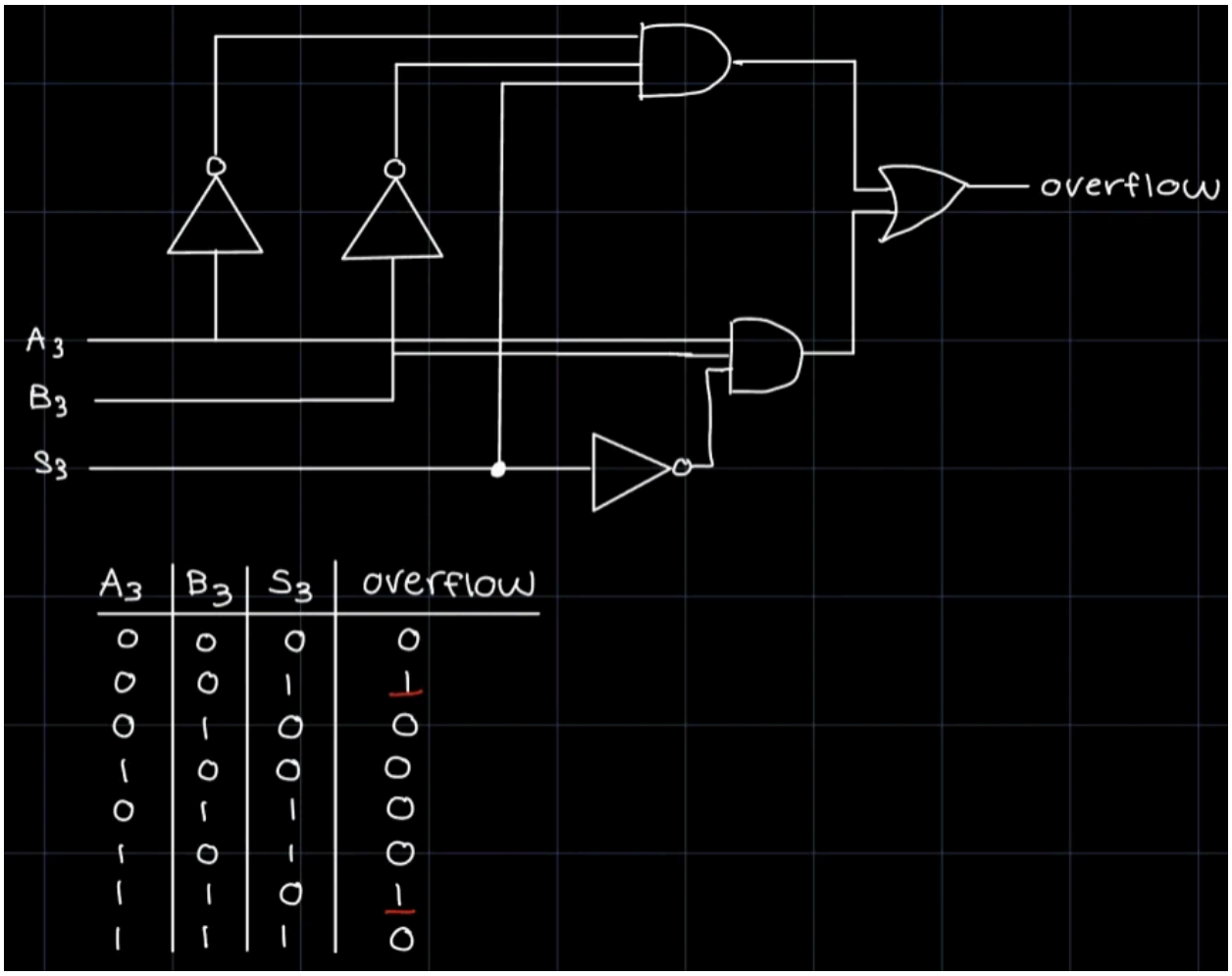


Figure 2. Overflow Detecting Unit